Namespace BOOSEapp

Classes

AppCanvas

Custom canvas implementation used by the BOOSE interpreter. Responsible for drawing shapes and managing the drawing cursor.

AppCommandFactory

Custom command factory used to create BOOSE commands. It extends the default command factory and returns application-specific command implementations.

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Bootstrap the application configuration.

Circle

BOOSE command that draws a circle on the canvas.

Form1

Main user interface for the BOOSE interpreter. Allows the user to input BOOSE programs, execute them, and display graphical output on the canvas.

MoveTo

BOOSE command that moves the drawing cursor to a specified position on the canvas.

Program

Rectangle

BOOSE command that draws a rectangle on the canvas.

Class AppCanvas

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Custom canvas implementation used by the BOOSE interpreter. Responsible for drawing shapes and managing the drawing cursor.

```
public class AppCanvas : ICanvas
```

Inheritance

object ← AppCanvas

Implements

ICanvas

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AppCanvas(int, int)

Initializes a new drawing canvas with the specified size.

```
public AppCanvas(int xsize, int ysize)
```

Parameters

`xsize` int

Width of the canvas.

`ysize` int

Height of the canvas.

Fields

CanvasBitmap

```
private Bitmap CanvasBitmap
```

Field Value

Bitmap

Pen

```
private Pen Pen
```

Field Value

Pen

g

```
private Graphics g
```

Field Value

Graphics

xPos

```
private int xPos
```

Field Value

int

yPos

```
private int yPos
```

Field Value

int

Properties

PenColour

Gets or sets the current pen colour.

```
public object PenColour { get; set; }
```

Property Value

object

Xpos

Gets or sets the current X position of the drawing cursor.

```
public int Xpos { get; set; }
```

Property Value

int

Ypos

Gets or sets the current Y position of the drawing cursor.

```
public int Ypos { get; set; }
```

Property Value

int

Methods

Circle(int, bool)

Draws a circle from the current cursor position.

```
public void Circle(int radius, bool filled)
```

Parameters

`radius` int

Radius of the circle.

`filled` bool

Determines whether the circle is filled.

Clear()

Clears the canvas to white.

```
public void Clear()
```

DrawTo(int, int)

Draws a line from the current cursor position to the given coordinates.

```
public void DrawTo(int x, int y)
```

Parameters

`x` int

Target X coordinate.

`y` int

Target Y coordinate.

MoveTo(int, int)

Moves the cursor to a new position without drawing.

```
public void MoveTo(int x, int y)
```

Parameters

`x` int

New X coordinate.

`y` int

New Y coordinate.

Rect(int, int, bool)

Draws a rectangle from the current cursor position.

```
public void Rect(int width, int height, bool filled)
```

Parameters

`width` int

Width of the rectangle.

`height` int

Height of the rectangle.

`filled` bool

Determines whether the rectangle is filled.

Reset()

Resets the cursor to default position.

```
public void Reset()
```

Set(int, int)

Resizes the canvas.

```
public void Set(int width, int height)
```

Parameters

`width` int

New width.

`height` int

New height.

SetColour(int, int, int)

Sets the drawing colour using RGB values.

```
public void SetColour(int red, int green, int blue)
```

Parameters

`red` int

Red component.

`green` int

Green component.

`blue` int

Blue component.

Tri(int, int)

Draws a triangle .

```
public void Tri(int width, int height)
```

Parameters

`width` int

`height` int

WriteText(string)

Writes text to the canvas.

```
public void WriteText(string text)
```

Parameters

`text` string

getBitmap()

Returns the bitmap representing the canvas.

```
public object getBitmap()
```

Returns

object

Bitmap object.

Class AppCommandFactory

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Custom command factory used to create BOOSE commands. It extends the default command factory and returns application-specific command implementations.

```
public class AppCommandFactory : CommandFactory, ICommandFactory
```

Inheritance

object ← CommandFactory ← AppCommandFactory

Implements

ICommandFactory

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Constructors

AppCommandFactory(ICanvas)

Initializes a new instance of the AppCommandFactory class.

```
public AppCommandFactory(ICanvas canvas)
```

Parameters

`canvas` ICanvas

Canvas used by commands created by the factory.

Fields

canvas

```
private ICanvas canvas
```

Field Value

ICanvas

Methods

MakeCommand(string)

Creates a command object based on the command name.

```
public override ICommand MakeCommand(string commandType)
```

Parameters

`commandType` string

The name of the BOOSE command.

Returns

ICommand

An BOOSE.ICommand implementation for the requested command.

Class ApplicationConfiguration

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Bootstrap the application configuration.

```
[CompilerGenerated]
internal static class ApplicationConfiguration
```

Inheritance

object ← ApplicationConfiguration

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Initialize()

Bootstrap the application as follows:

```
global::System.Windows.Forms.Application.EnableVisualStyles();
global::System.Windows.Forms.Application.SetCompatibleTextRenderingDefault(false);
global::System.Windows.Forms.Application.SetHighDpiMode(HighDpiMode.SystemAware);
```

```
public static void Initialize()
```

Class Circle

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

BOOSE command that draws a circle on the canvas.

```
internal class Circle : CommandOneParameter, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← Circle

Implements

ICommand

Inherited Members

CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CommandOneParameter.CheckParameters(string[]) , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Circle()

Default constructor used by the command factory.

```
public Circle()
```

Circle(Canvas, int)

Initializes a new instance of the Circle command.

```
public Circle(Canvas c, int radius)
```

Parameters

`c` Canvas

Canvas used for drawing.

`radius` int

Radius of the circle.

Fields

radius

```
private int radius
```

Field Value

int

Methods

Execute()

Executes the circle command and draws the circle.

```
public override void Execute()
```

Exceptions

CanvasException

Thrown when the radius parameter is invalid.

Class Form1

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

Main user interface for the BOOSE interpreter. Allows the user to input BOOSE programs, execute them, and display graphical output on the canvas.

```
public class Form1 : Form, IOleControl.Interface, IOleObject.Interface,
IOleInPlaceObject.Interface, IOleInPlaceActiveObject.Interface, IOleWindow.Interface,
IViewObject2.Interface, IViewObject.Interface, IPersistStreamInit.Interface,
IPersistPropertyBag.Interface, IPersistStorage.Interface, IPersist.Interface,
IQuickActivate.Interface, ISupportOleDropSource, IDropTarget, ISynchronizeInvoke,
IWin32Window, IBindableComponent, IKeyboardToolTip, IHandle<HWND>, IArrangedElement,
IComponent, IDisposable, IContainerControl
```

Inheritance

object ← MarshalByRefObject ← Component ← Control ← ScrollableControl ← ContainerControl ← Form ← Form1

Implements

IOleControl.Interface, IOleObject.Interface, IOleInPlaceObject.Interface, IOleInPlaceActiveObject.Interface, IOleWindow.Interface, IViewObject2.Interface, IViewObject.Interface, IPersistStreamInit.Interface, IPersistPropertyBag.Interface, IPersistStorage.Interface, IPersist.Interface, IQuickActivate.Interface, ISupportOleDropSource, IDropTarget, ISynchronizeInvoke, IWin32Window, IBindableComponent, IKeyboardToolTip, IHandle<HWND>, IArrangedElement, IComponent, IDisposable, IContainerControl

Inherited Members

Form.SetVisibleCore(bool) , Form.OnFormCornerPreferenceChanged(EventArgs) , Form.OnFormBorderColorChanged(EventArgs) , Form.OnFormCaptionBackColorChanged(EventArgs) , Form.OnFormCaptionTextColorChanged(EventArgs) , Form.Activate() , Form.ActivateMdiChild(Form) , Form.AddOwnedForm(Form) , Form.AdjustFormScrollbars(bool) , Form.ApplyAutoScaling() , Form.AssignParent(Control) , Form.Close() , Form.CreateAccessibilityInstance() , Form.CreateControlsInstance() , Form.AfterControlRemoved(Control, Control) , Form.CreateHandle() , Form.DefWndProc(ref Message) , Form.FocusInternal() , Form.GetAutoScaleSize(Font) , Form.CanSelectCore() , Form.CanProcessMnemonic() , Form.ProcessMnemonic(char) , Form.CenterToParent() , Form.CenterToScreen() , Form.LayoutMdi(MdiLayout) ,

Form.OnActivated(EventArgs) , Form.OnAutoScaleModeChanged() ,
Form.OnBackgroundImageChanged(EventArgs) ,
Form.OnBackgroundImageLayoutChanged(EventArgs) , Form.OnClosing(CancelEventArgs) ,
Form.OnClosed(EventArgs) , Form.OnFormClosing(FormClosingEventArgs) ,
Form.OnFormClosed(FormClosedEventArgs) , Form.OnCreateControl() ,
Form.OnDeactivate(EventArgs) , Form.OnEnabledChanged(EventArgs) , Form.OnEnter(EventArgs) ,
Form.OnFontChanged(EventArgs) , Form.OnGotFocus(EventArgs) ,
Form.OnHandleCreated(EventArgs) , Form.OnHandleDestroyed(EventArgs) ,
Form.OnHelpButtonClicked(CancelEventArgs) , Form.OnLayout(LayoutEventArgs) ,
Form.OnLoad(EventArgs) , Form.OnMaximizedBoundsChanged(EventArgs) ,
Form.OnMaximumSizeChanged(EventArgs) , Form.OnMinimumSizeChanged(EventArgs) ,
Form.OnInputLanguageChanged(InputLanguageChangedEventArgs) ,
Form.OnInputLanguageChanging(InputLanguageChangingEventArgs) ,
Form.OnVisibleChanged(EventArgs) , Form.OnMdiChildActivate(EventArgs) ,
Form.OnMenuStart(EventArgs) , Form.OnMenuComplete(EventArgs) ,
Form.OnPaint(PaintEventArgs) , Form.OnResize(EventArgs) ,
Form.OnDpiChanged(DpiChangedEventArgs) , Form.OnGetDpiScaledSize(int, int, ref Size) ,
Form.OnRightToLeftLayoutChanged(EventArgs) , Form.OnShown(EventArgs) ,
Form.OnTextChanged(EventArgs) , Form.ProcessCmdKey(ref Message, Keys) ,
Form.ProcessDialogKey(Keys) , Form.ProcessDialogChar(char) ,
Form.ProcessKeyPreview(ref Message) , Form.ProcessTabKey(bool) , Form.RecreateHandleCore() ,
Form.RemoveOwnedForm(Form) , Form.Select(bool, bool) , Form.ScaleCore(float, float) ,
Form.ScaleMinMaxSize(float, float, bool) ,
Form.GetScaledBounds(Rectangle, SizeF, BoundsSpecified) ,
Form.ScaleControl(SizeF, BoundsSpecified) , Form.SetBoundsCore(int, int, int, int, BoundsSpecified) ,
Form.SetClientSizeCore(int, int) , Form.SetDesktopBounds(int, int, int, int) ,
Form.SetDesktopLocation(int, int) , Form.Show(IWin32Window) ,
Form.ShowAsync(IWin32Window) , Form.SuppressApplicationOnThreadException(Exception) ,
Form.ShowDialog() , Form.ShowDialog(IWin32Window) , Form.ShowDialogAsync() ,
Form.ShowDialogAsync(IWin32Window) , Form.ShouldSerializeAutoScaleBaseSize() ,
Form.ShouldSerializeSize() , Form.ToString() , Form.UpdateDefaultButton() ,
Form.OnResizeBegin(EventArgs) , Form.OnResizeEnd(EventArgs) ,
Form.OnStyleChanged(EventArgs) , Form.ValidateChildren() ,
Form.ValidateChildren(ValidationConstraints) , Form.WndProc(ref Message) , Form.AcceptButton ,
Form.ActiveForm , Form.ActiveMdiChild , Form.AllowTransparency , Form.AutoScale ,
Form.AutoScaleBaseSize , Form.AutoScroll , Form.AutoSize , Form.AutoSizeMode ,
Form.AutoValidate , Form.BackColor , Form.FormBorderStyle , Form.CancelButton ,
Form.ClientSize , Form.ControlBox , Form.CreateParams , Form.DefaultImeMode ,
Form.DefaultSize , Form.DesktopBounds , Form.DesktopLocation , Form.DialogResult ,
Form.HelpButton , Form.Icon , Form.IsMdiChild , Form.IsMdiContainer ,

Form.IsRestrictedWindow , Form.KeyPreview , Form.Location , Form.MaximizedBounds , Form.MaximumSize , Form.MainMenuStrip , Form.Margin , Form.MinimumSize , Form.MaximizeBox , Form.MdiChildren , Form.MdiChildrenMinimizedAnchorBottom , Form.MdiParent , Form.MinimizeBox , Form.Modal , Form.Opacity , Form.OwnedForms , Form.Owner , Form.RestoreBounds , Form.RightToLeftLayout , Form.ParentInternal , Form.ShowInTaskbar , Form.ShowIcon , Form.ShowParams , Form.ShowWithoutActivation , Form.Size , Form.SizeGripStyle , Form.StartPosition , Form.TabIndex , Form.TabStop , Form.Text , Form.TopLevel , Form.TopMost , Form.TransparencyKey , Form.FormCornerPreference , Form.FormBorderColor , Form.FormCaptionBackColor , Form.FormCaptionTextColor , Form.WindowState , Form.WindowText , Form.SupportsUiaProviders , Form.AutoSizeChanged , Form.AutoValidateChanged , Form.HelpButtonClicked , Form.MaximizedBoundsChanged , Form.MaximumSizeChanged , Form.MarginChanged , Form.MinimumSizeChanged , Form.TabIndexChanged , Form.TabStopChanged , Form.Activated , Form.Closing , Form.Closed , Form.Deactivate , Form.FormClosing , Form.FormBorderColorChanged , Form.FormCaptionBackColorChanged , Form.FormCaptionTextColorChanged , Form.FormCornerPreferenceChanged , Form.FormClosed , Form.Load , Form.MdiChildActivate , Form.MenuComplete , Form.MenuStart , Form.InputLanguageChanged , Form.InputLanguageChanging , Form.RightToLeftLayoutChanged , Form.Shown , Form.DpiChanged , Form.ResizeBegin , Form.ResizeEnd , ContainerControl.IContainerControl.ActivateControl(Control) , ContainerControl.GetPreferredSizeCore(Size) , ContainerControl.GetToolNativeScreenRectangle() , ContainerControl.OnAutoValidateChanged(EventArgs) , ContainerControl.OnFrameWindowActivate(bool) , ContainerControl.OnChildLayoutResuming(Control, bool) , ContainerControl.OnLayoutResuming(bool) , ContainerControl.OnMove(EventArgs) , ContainerControl.OnParentChanged(EventArgs) , ContainerControl.PerformAutoScale() , ContainerControl.Scale(SizeF, SizeF, Control, bool) , ContainerControl.RescaleConstantsForDpi(int, int) , ContainerControl.Validate() , ContainerControl.Validate(bool) , ContainerControl.ShouldSerializeAutoValidate() , ContainerControl.AutoScaleDimensions , ContainerControl.AutoScaleFactor , ContainerControl.AutoScaleMode , ContainerControl.BindingContext , ContainerControl.CanEnableIme , ContainerControl.ActiveControl , ContainerControl.CurrentAutoScaleDimensions , ContainerControl.ParentForm , ContainerControl.InnerMostActiveContainerControl , ScrollableControl.ScrollStateAutoScrolling , ScrollableControl.ScrollStateHScrollVisible , ScrollableControl.ScrollStateVScrollVisible , ScrollableControl.ScrollStateUserHasScrolled , ScrollableControl.ScrollStateFullDrag , ScrollableControl.GetScrollState(int) , ScrollableControl.OnMouseWheel(MouseEventArgs) , ScrollableControl.OnRightToLeftChanged(EventArgs) , ScrollableControl.OnPaintBackground(PaintEventArgs) , ScrollableControl.OnPaddingChanged(EventArgs) , ScrollableControl.SetDisplayRectLocation(int, int) ,

ScrollableControl.ScrollControlIntoView(Control) , ScrollableControl.ScrollToControl(Control) ,
ScrollableControl.OnScroll(ScrollEventArgs) , ScrollableControl.SetAutoScrollMargin(int, int) ,
ScrollableControl.SetScrollState(int, bool) , ScrollableControl.AutoScrollMargin ,
ScrollableControl.AutoScrollPosition , ScrollableControl.AutoScrollMinSize ,
ScrollableControl.DisplayRectangle , ScrollableControl.IArrangedElement.DisplayRectangle ,
ScrollableControl.HScroll , ScrollableControl.HorizontalScroll , ScrollableControl.VScroll ,
ScrollableControl.VerticalScroll , ScrollableControl.DockPadding , ScrollableControl.Scroll ,
Control.s_paddingChangedEvent , Control.IOleControl.Interface.GetControlInfo(CONTROLINFO*) ,
Control.IOleControl.Interface.OnMnemonic(MSG*) ,
Control.IOleControl.Interface.OnAmbientPropertyChange(int) ,
Control.IOleControl.Interface.FreezeEvents(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.GetWindow(HWND*) ,
Control.IOleInPlaceActiveObject.Interface.ContextSensitiveHelp(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.TranslateAccelerator(MSG*) ,
Control.IOleInPlaceActiveObject.Interface.OnFrameWindowActivate(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.OnDocWindowActivate(BOOL) ,
Control.IOleInPlaceActiveObject.Interface.ResizeBorder(RECT*, IOleInPlaceUIWindow*, BOOL) ,
Control.IOleInPlaceActiveObject.Interface.EnableModeless(BOOL) ,
Control.IOleInPlaceObject.Interface.GetWindow(HWND*) ,
Control.IOleInPlaceObject.Interface.ContextSensitiveHelp(BOOL) ,
Control.IOleInPlaceObject.Interface.InPlaceDeactivate() ,
Control.IOleInPlaceObject.Interface.UIDeactivate() ,
Control.IOleInPlaceObject.Interface.SetObjectRects(RECT*, RECT*) ,
Control.IOleInPlaceObject.Interface.ReactivateAndUndo() ,
Control.IOleObject.Interface.SetClientSite(IOleClientSite*) ,
Control.IOleObject.Interface.GetClientSite(IOleClientSite**) ,
Control.IOleObject.Interface.SetHostNames(PCWSTR, PCWSTR) ,
Control.IOleObject.Interface.Close(uint) , Control.IOleObject.Interface.SetMoniker(uint, IMoniker*) ,
Control.IOleObject.Interface.GetMoniker(uint, uint, IMoniker**) ,
Control.IOleObject.Interface.InitFromData(IDataObject*, BOOL, uint) ,
Control.IOleObject.Interface.GetClipboardData(uint, IDataObject**) ,
Control.IOleObject.Interface.DoVerb(int, MSG*, IOleClientSite*, int, HWND, RECT*) ,
Control.IOleObject.Interface.EnumVerbs(IEnumOLEVERB**) , Control.IOleObject.Interface.Update() ,
Control.IOleObject.Interface.IsUpToDate() , Control.IOleObject.Interface.GetUserClassID(Guid*) ,
Control.IOleObject.Interface.GetUserType(uint, PWSTR*) ,
Control.IOleObject.Interface.SetExtent(DVASPECT, SIZE*) ,
Control.IOleObject.Interface.GetExtent(DVASPECT, SIZE*) ,
Control.IOleObject.Interface.Advise(IAdviseSink*, uint*) , Control.IOleObject.Interface.Unadvise(uint) ,
Control.IOleObject.Interface.EnumAdvise(IEnumSTATDATA**) ,
Control.IOleObject.Interface.GetMiscStatus(DVASPECT, OLEMISC*) ,

Control.IOleObject.Interface.SetColorScheme(LOGPALETTE*) ,
Control.IOleWindow.Interface.GetWindow(HWND*) ,
Control.IOleWindow.Interface.ContextSensitiveHelp(BOOL) ,
Control.IPersist.Interface.GetClassID(Guid*) , Control.IPersistPropertyBag.Interface.InitNew() ,
Control.IPersistPropertyBag.Interface.GetClassID(Guid*) ,
Control.IPersistPropertyBag.Interface.Load(IPropertyBag*, IErrorLog*) ,
Control.IPersistPropertyBag.Interface.Save(IPropertyBag*, BOOL, BOOL) ,
Control.IPersistStorage.Interface.GetClassID(Guid*) , Control.IPersistStorage.Interface.IsDirty() ,
Control.IPersistStorage.Interface.InitNew(IStorage*) ,
Control.IPersistStorage.Interface.Load(IStorage*) ,
Control.IPersistStorage.Interface.Save(IStorage*, BOOL) ,
Control.IPersistStorage.Interface.SaveCompleted(IStorage*) ,
Control.IPersistStorage.Interface.HandsOffStorage() ,
Control.IPersistStreamInit.Interface.GetClassID(Guid*) , Control.IPersistStreamInit.Interface.IsDirty() ,
Control.IPersistStreamInit.Interface.Load(IStream*) ,
Control.IPersistStreamInit.Interface.Save(IStream*, BOOL) ,
Control.IPersistStreamInit.Interface.GetSizeMax(ulong*) ,
Control.IPersistStreamInit.Interface.InitNew() ,
Control.IQuickActivate.Interface.QuickActivate(QACONTAINER*, QACONTROL*) ,
Control.IQuickActivate.Interface.SetContentExtent(SIZE*) ,
Control.IQuickActivate.Interface.GetContentExtent(SIZE*) ,
Control.IViewObject.Interface.Draw(DVASPECT, int, void*, DVTARGETDEVICE*, HDC, HDC, RECTL*, RECTL*,
nint, nuint) ,
Control.IViewObject.Interface.GetColorSet(DVASPECT, int, void*, DVTARGETDEVICE*, HDC,
LOGPALETTE**) ,
Control.IViewObject.Interface.Freeze(DVASPECT, int, void*, uint*) ,
Control.IViewObject.Interface.Unfreeze(uint) ,
Control.IViewObject.Interface.SetAdvise(DVASPECT, uint, IAdviseSink*) ,
Control.IViewObject.Interface.GetAdvise(uint*, uint*, IAdviseSink**) ,
Control.IViewObject2.Interface.Draw(DVASPECT, int, void*, DVTARGETDEVICE*, HDC, HDC, RECTL*,
RECTL*, nint, nuint) ,
Control.IViewObject2.Interface.GetColorSet(DVASPECT, int, void*, DVTARGETDEVICE*, HDC,
LOGPALETTE**) ,
Control.IViewObject2.Interface.Freeze(DVASPECT, int, void*, uint*) ,
Control.IViewObject2.Interface.Unfreeze(uint) ,
Control.IViewObject2.Interface.SetAdvise(DVASPECT, uint, IAdviseSink*) ,
Control.IViewObject2.Interface.GetAdvise(uint*, uint*, IAdviseSink**) ,
Control.IViewObject2.Interface.GetExtent(DVASPECT, int, DVTARGETDEVICE*, SIZE*) ,
Control.GetAccessibilityObjectById(int) , Control.SetAutoSizeMode(AutoSizeMode) ,
Control.GetAutoSizeMode() , Control.ResetBindings() , Control.GetPreferredSize(Size) ,

Control.IsHoveredWithMouse() , Control.AddReflectChild() , Control.RemoveReflectChild() ,
Control.RemoveToolTip(ToolTip) , Control.AccessibilityNotifyClients(AccessibleEvents, int) ,
Control.AccessibilityNotifyClients(AccessibleEvents, int, int) , Control.BeginInvoke(Delegate) ,
Control.BeginInvoke(Action) , Control.BeginInvoke(Delegate, params object[]) ,
Control.BringToFront() , Control.Contains(Control) , Control.CreateGraphics() ,
Control.CreateControl() , Control.DestroyHandle() , Control.DisposeAxControls() ,
Control.DoDragDrop(object, DragDropEffects) ,
Control.DoDragDrop(object, DragDropEffects, Bitmap, Point, bool) ,
Control.DrawToBitmap(Bitmap, Rectangle) , Control.EndInvoke(IAsyncResult) , Control.FindForm() ,
Control.GetTopLevel() , Control.RaiseKeyEvent(object, KeyEventArgs) ,
Control.RaiseMouseEvent(object, MouseEventArgs) , Control.Focus() ,
Control.FromChildHandle(nint) , Control.FromHandle(nint) ,
Control.ApplyBoundsConstraints(int, int, int, int) ,
Control.GetChildAtPoint(Point, GetChildAtPointSkip) , Control.GetCaptionForTool(ToolTip) ,
Control.GetChildAtPoint(Point) , Control.GetContainerControl() ,
Control.GetFirstChildControlInTabOrder(bool) , Control.GetNeighboringToolsRectangles() ,
Control.GetNextControl(Control, bool) , Control.GetState(Control.States) ,
Control.GetExtendedState(Control.ExtendedStates) , Control.GetStyle(ControlStyles) ,
Control.Hide() , Control.InitLayout() , Control.InitializeDCForWmCtlColor(HDC, MessageId) ,
Control.Invalidate(Region) , Control.Invalidate(Region, bool) , Control.Invalidate() ,
Control.Invalidate(bool) , Control.Invalidate(Rectangle) , Control.Invalidate(Rectangle, bool) ,
Control.Invoke(Action) , Control.Invoke(Delegate) , Control.Invoke(Delegate, params object[]) ,
Control.Invoke<T>(Func<T>) , Control.InvokePaint(Control, PaintEventArgs) ,
Control.InvokePaintBackground(Control, PaintEventArgs) , Control.IsKeyLocked(Keys) ,
Control.IsInputChar(char) , Control.IsInputKey(Keys) , Control.IsMnemonic(char, string) ,
Control.LogicalToDeviceUnits(int) , Control.LogicalToDeviceUnits(Size) ,
Control.ScaleBitmapLogicalToDevice(ref Bitmap) ,
Control.AdjustWindowRectExForControlDpi(ref RECT, WINDOW_STYLE, bool, WINDOW_EX_STYLE) ,
Control.NotifyInvalidate(Rectangle) , Control.InvokeOnClick(Control, EventArgs) ,
Control.OnAutoSizeChanged(EventArgs) , Control.OnBackColorChanged(EventArgs) ,
Control.OnBindingContextChanged(EventArgs) , Control.OnCausesValidationChanged(EventArgs) ,
Control.OnContextMenuStripChanged(EventArgs) , Control.OnCursorChanged(EventArgs) ,
Control.OnDataContextChanged(EventArgs) , Control.OnDockChanged(EventArgs) ,
Control.OnForeColorChanged(EventArgs) , Control.OnNotifyMessage(Message) ,
Control.OnParentBackColorChanged(EventArgs) ,
Control.OnParentBackgroundImageChanged(EventArgs) ,
Control.OnParentBindingContextChanged(EventArgs) , Control.OnParentCursorChanged(EventArgs) ,
Control.OnParentDataContextChanged(EventArgs) , Control.OnParentEnabledChanged(EventArgs) ,
Control.OnParentFontChanged(EventArgs) , Control.OnParentHandleRecreated() ,
Control.OnParentHandleRecreating() , Control.OnParentForeColorChanged(EventArgs) ,

Control.OnParentRightToLeftChanged(EventArgs) , Control.OnParentVisibleChanged(EventArgs) ,
Control.OnParentBecameInvisible() , Control.OnPrint(PaintEventArgs) ,
Control.OnTabIndexChanged(EventArgs) , Control.OnTabStopChanged(EventArgs) ,
Control.OnTopMostActiveXParentChanged(EventArgs) , Control.OnClick(EventArgs) ,
Control.OnClientSizeChanged(EventArgs) , Control.OnControlAdded(ControlEventArgs) ,
Control.OnControlRemoved(ControlEventArgs) ,
Control.OnInvokedSetScrollPosition(object, EventArgs) , Control.OnLocationChanged(EventArgs) ,
Control.OnDoubleClick(EventArgs) , Control.OnDragEnter(DragEventArgs) ,
Control.OnDragOver(DragEventArgs) , Control.OnDragLeave(EventArgs) ,
Control.OnDragDrop(DragEventArgs) , Control.OnGiveFeedback(GiveFeedbackEventArgs) ,
Control.InvokeGotFocus(Control, EventArgs) , Control.OnHelpRequested(HelpEventArgs) ,
Control.OnInvalidated(InvalidateEventArgs) , Control.OnKeyDown(KeyEventArgs) ,
Control.OnKeyPress(KeyPressEventArgs) , Control.OnKeyUp(KeyEventArgs) ,
Control.OnLayoutSuspended() , Control.OnLeave(EventArgs) ,
Control.InvokeLostFocus(Control, EventArgs) , Control.OnLostFocus(EventArgs) ,
Control.OnMarginChanged(EventArgs) , Control.OnMouseDoubleClick(MouseEventArgs) ,
Control.OnMouseClick(MouseEventArgs) , Control.OnMouseCaptureChanged(EventArgs) ,
Control.OnMouseDown(MouseEventArgs) , Control.OnMouseEnter(EventArgs) ,
Control.OnMouseLeave(EventArgs) , Control.OnDpiChangedBeforeParent(EventArgs) ,
Control.OnDpiChangedAfterParent(EventArgs) , Control.OnMouseHover(EventArgs) ,
Control.OnMouseMove(MouseEventArgs) , Control.OnMouseUp(MouseEventArgs) ,
Control.OnQueryContinueDrag(QueryContinueDragEventArgs) ,
Control.OnRegionChanged(EventArgs) , Control.OnPreviewKeyDown(PreviewKeyDownEventArgs) ,
Control.OnSizeChanged(EventArgs) , Control.OnChangeUICues(UICuesEventArgs) ,
Control.OnSystemColorsChanged(EventArgs) , Control.OnValidating(CancelEventArgs) ,
Control.OnValidated(EventArgs) , Control.InitializeConstantsForInitialDpi(int) ,
Control.PerformLayout() , Control.PerformLayout(Control, string) , Control.PointToClient(Point) ,
Control.PointToScreen(Point) , Control.PreProcessMessage(ref Message) ,
Control.PreProcessControlMessage(ref Message) ,
Control.PrintToMetaFileRecursive(HDC, nint, Rectangle) , Control.ProcessKeyEventArgs(ref Message) ,
Control.ProcessKeyMessage(ref Message) , Control.RaiseDragEvent(object, DragEventArgs) ,
Control.RaisePaintEvent(object, PaintEventArgs) , Control.ResetBackColor() , Control.ResetCursor() ,
Control.ResetFont() , Control.ResetForeColor() , Control.ResetRightToLeft() ,
Control.RecreateHandle() , Control.RectangleToClient(Rectangle) ,
Control.RectangleToScreen(Rectangle) , Control.ReflectMessage(nint, ref Message) ,
Control.Refresh() , Control.ReleaseUiaProvider(HWND) ,
Control.TryGetAccessibilityObject(out AccessibleObject) , Control.ResetMouseEventArgs() ,
Control.ResetText() , Control.ResumeLayout() , Control.ResumeLayout(bool) , Control.Scale(float) ,
Control.Scale(float, float) , Control.Scale(SizeF) , Control.Select() ,
Control.SelectNextControl(Control, bool, bool, bool, bool) , Control.SendToBack() ,

Control.SetBounds(int, int, int, int) , Control.SetBounds(int, int, int, int, BoundsSpecified) ,
Control.SizeFromClientSize(Size) , Control.SetState(Control.States, bool) ,
Control.SetExtendedState(Control.ExtendedStates, bool) , Control.SetStyle(ControlStyles, bool) ,
Control.SetToolTip(ToolTip) , Control.SetTopLevel(bool) , Control.SetTopLevelInternal(bool) ,
Control.ShouldPerformContainerValidation() , Control.ShouldSerializeBackColor() ,
Control.ShouldSerializeCursor() , Control.ShouldSerializeForeColor() , Control.ShouldSerializeFont() ,
Control.ShouldSerializeRightToLeft() , Control.RtlTranslateAlignment(HorizontalAlignment) ,
Control.RtlTranslateAlignment(LeftRightAlignment) ,
Control.RtlTranslateAlignment(ContentAlignment) ,
Control.RtlTranslateHorizontal(HorizontalAlignment) ,
Control.RtlTranslateLeftRight(LeftRightAlignment) , Control.RtlTranslateContent(ContentAlignment) ,
Control.Show() , Control.ShouldSerializeMaximumSize() , Control.ShouldSerializeMinimumSize() ,
Control.ShouldSerializeText() , Control.SuspendLayout() , Control.TryGetExplicitlySetFont(out Font) ,
Control.SetScaledFont(Font, bool) , Control.Update() , Control.UpdateBounds() ,
Control.UpdateBounds(int, int, int, int) , Control.UpdateBounds(int, int, int, int, int, int) ,
Control.UpdateZOrder() , Control.UpdateStyles() , Control.UpdateStylesCore() ,
Control.OnBoundsUpdate(int, int, int, int) , Control.WmContextMenu(ref Message) ,
Control.IArrangedElement.PerformLayout(IArrangedElement, string) ,
Control.IArrangedElement.SetBounds(Rectangle, BoundsSpecified) ,
Control.IDropTarget.OnDragEnter(DragEventArgs) ,
Control.IDropTarget.OnDragOver(DragEventArgs) , Control.IDropTarget.OnDragLeave(EventArgs) ,
Control.IDropTarget.OnDragDrop(DragEventArgs) ,
Control.ISupportOleDropSource.OnGiveFeedback(GiveFeedbackEventArgs) ,
Control.ISupportOleDropSource.OnQueryContinueDrag(QueryContinueDragEventArgs) ,
Control.IKeyboardToolTip.CanShowToolTipsNow() ,
Control.IKeyboardToolTip.GetNativeScreenRectangle() ,
Control.IKeyboardToolTip.GetNeighboringToolsRectangles() ,
Control.IKeyboardToolTip.IsHoveredWithMouse() , Control.IKeyboardToolTip.HasRtlModeEnabled() ,
Control.IKeyboardToolTip.AllowsToolTip() , Control.IKeyboardToolTip.GetOwnerWindow() ,
Control.IKeyboardToolTip.OnHooked(ToolTip) , Control.IKeyboardToolTip.OnUnhooked(ToolTip) ,
Control.IKeyboardToolTip.GetCaptionForTool(ToolTip) ,
Control.IKeyboardToolTip.ShowsOwnToolTip() , Control.IKeyboardToolTip.IsBeingTabbedTo() ,
Control.IKeyboardToolTip.AllowsChildrenToShowToolTips() , Control.ShowsOwnKeyboardToolTip() ,
Control.OnKeyboardToolTipHook(ToolTip) , Control.OnKeyboardToolTipUnhook(ToolTip) ,
Control.AllowsKeyboardToolTip() , Control.GetToolInfoWrapper(TOOLTIP_FLAGS, string, ToolTip) ,
Control.AllowsChildrenToShowToolTips() , Control.OnImeModeChanged(EventArgs) ,
Control.ResetImeMode() , Control.ShouldSerializeImeMode() ,
Control.InvokeAsync(Action, CancellationToken) ,
Control.InvokeAsync<T>(Func<T>, CancellationToken) ,
Control.InvokeAsync(Func<CancellationToken, ValueTask>, CancellationToken) ,

Control.InvokeAsync<T>(Func<CancellationToken, ValueTask<T>>, CancellationToken) ,
Control.AccessibilityObject , Control.AccessibleDefaultActionDescription ,
Control.AccessibleDescription , Control.AccessibleName , Control.AccessibleRole ,
Control.AllowDrop , Control.Anchor , Control.AutoScrollOffset , Control.LayoutEngine ,
Control.DataContext , Control.BackgroundImage , Control.BackgroundImageLayout ,
Control.Bottom , Control.Bounds , Control.CanAccessProperties , Control.CanFocus ,
Control.CanRaiseEvents , Control.CanSelect , Control.Capture , Control.CausesValidation ,
Control.CheckForIllegalCrossThreadCalls , Control.ClientRectangle , Control.CompanyName ,
Control.ContainsFocus , Control.ContextMenuStrip , Control.Controls , Control.Created ,
Control.Cursor , Control.DataBindings , Control.DefaultBackColor , Control.DefaultCursor ,
Control.DefaultFont , Control.DefaultForeColor , Control.DefaultMargin ,
Control.DefaultMaximumSize , Control.DefaultMinimumSize , Control.DefaultPadding ,
Control.DeviceDpi , Control.IsDisposed , Control.Disposing , Control.Dock ,
Control.DoubleBuffered , Control.Enabled , Control.Focused , Control.Font ,
Control.FontHeight , Control.ForeColor , Control.Handle , Control.HasChildren , Control.Height ,
Control.IsHandleCreated , Control.InvokeRequired , Control.IsAccessible ,
Control.IsAncestorSiteInDesignMode , Control.IsContainerControl , Control.IsMirrored ,
Control.IsMnemonicsListenerAxSourced , Control.LastKeyData , Control.Left ,
Control.ModifierKeys , Control.MouseButtons , Control.MousePosition , Control.Name ,
Control.Parent , Control.ProductName , Control.ProductVersion , Control.RecreatingHandle ,
Control.Region , Control.RenderRightToLeft , Control.RenderTransparencyWithVisualStyles ,
Control.ResizeRedraw , Control.Right , Control.RightToLeft , Control.ScaleChildren , Control.Site ,
Control.Tag , Control.Top , Control.TopLevelControl , Control.ShowKeyboardCues ,
Control.ShowFocusCues , Control.UseWaitCursor , Control.SupportsUseCompatibleTextRendering ,
Control.Visible , Control.Width , Control.ExtendedWindowStyle , Control.WindowTarget ,
Control.PreferredSize , Control.Padding , Control.IArrangedElement.Children ,
Control.IArrangedElement.Container , Control.IArrangedElement.ParticipatesInLayout ,
Control.IArrangedElement.Properties , Control.IHandle<HWND>.Handle , Control.ImeMode ,
Control.ImeModeBase , Control.PropagatingImeMode , Control.BackColorChanged ,
Control.BackgroundImageChanged , Control.BackgroundImageLayoutChanged ,
Control.BindingContextChanged , Control.CausesValidationChanged , Control.ClientSizeChanged ,
Control.ContextMenuStripChanged , Control.CursorChanged , Control.DockChanged ,
Control.EnabledChanged , Control.FontChanged , Control.ForeColorChanged ,
Control.LocationChanged , Control.RegionChanged , Control.RightToLeftChanged ,
Control.SizeChanged , Control.TextChanged , Control.VisibleChanged , Control.Click ,
Control.ControlAdded , Control.ControlRemoved , Control.DataContextChanged ,
Control.DragDrop , Control.DragEnter , Control.DragOver , Control.DragLeave ,
Control.GiveFeedback , Control.HandleCreated , Control.HandleDestroyed ,
Control.HelpRequested , Control.Invalidated , Control.PaddingChanged , Control.Paint ,
Control.QueryContinueDrag , Control.QueryAccessibilityHelp , Control.DoubleClick ,

Control.Enter , Control.GotFocus , Control.KeyDown , Control.KeyPress , Control.KeyUp , Control.Layout , Control.Leave , Control.LostFocus , Control.MouseClick , Control.MouseDoubleClick , Control.MouseCaptureChanged , Control.MouseDown , Control.MouseEnter , Control.MouseLeave , Control.DpiChangedBeforeParent , Control.DpiChangedAfterParent , Control.MouseHover , Control.MouseMove , Control.MouseUp , Control.MouseWheel , Control.Move , Control.PreviewKeyDown , Control.Resize , Control.ChangeUICues , Control.StyleChanged , Control.SystemColorsChanged , Control.Validating , Control.Validated , Control.ParentChanged , Control.ImeModeChanged , Component.Dispose() , Component.GetService(Type) , Component.Container , Component.DesignMode , Component.Events , Component.Disposed , MarshalByRefObject.GetLifetimeService() , MarshalByRefObject.InitializeLifetimeService() , MarshalByRefObject.MemberwiseClone(bool) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Form1()

Initializes the main form and sets up BOOSE components.

```
public Form1()
```

Fields

Factory

```
private CommandFactory Factory
```

Field Value

CommandFactory

Parser

```
private IParser Parser
```

Field Value

IParser

Program

```
private StoredProgram Program
```

Field Value

StoredProgram

button1

```
private Button button1
```

Field Value

Button

canvas

```
private AppCanvas canvas
```

Field Value

AppCanvas

components

Required designer variable.

```
private IContainer components
```

Field Value

IContainer

pictureBox1

```
private PictureBox pictureBox1
```

Field Value

PictureBox

textBox1

```
private TextBox textBox1
```

Field Value

TextBox

Properties

UserInput

Gets or sets the BOOSE program entered by the user.

```
[Browsable(false)]
[DesignerSerializationVisibility(DesignerSerializationVisibility.Hidden)]
public string UserInput { get; set; }
```

Property Value

string

Methods

Dispose(bool)

Clean up any resources being used.

```
protected override void Dispose(bool disposing)
```

Parameters

`disposing` bool

true if managed resources should be disposed; otherwise, false.

Form1_Load(object, EventArgs)

```
private void Form1_Load(object sender, EventArgs e)
```

Parameters

`sender` object

`e` EventArgs

InitializeComponent()

Required method for Designer support - do not modify the contents of this method with the code editor.

```
private void InitializeComponent()
```

button1_Click(object, EventArgs)

```
public void button1_Click(object sender, EventArgs e)
```

Parameters

sender object

e EventArgs

panel1_Paint(object, PaintEventArgs)

```
private void panel1_Paint(object sender, PaintEventArgs e)
```

Parameters

sender object

e PaintEventArgs

pictureBox1_Click(object, EventArgs)

```
private void pictureBox1_Click(object sender, EventArgs e)
```

Parameters

sender object

e EventArgs

pictureBox1_Paint(object, PaintEventArgs)

```
private void pictureBox1_Paint(object sender, PaintEventArgs e)
```

Parameters

sender object

e PaintEventArgs

textBox1_TextChanged(object, EventArgs)

```
private void textBox1_TextChanged(object sender, EventArgs e)
```

Parameters

sender object

e EventArgs

textBox2_TextChanged(object, EventArgs)

```
private void textBox2_TextChanged(object sender, EventArgs e)
```

Parameters

sender object

e EventArgs

Class MoveTo

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

BOOSE command that moves the drawing cursor to a specified position on the canvas.

```
internal class MoveTo : CommandTwoParameters, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← MoveTo

Implements

ICommand

Inherited Members

CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed , CommandTwoParameters.CheckParameters(string[]) , CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

MoveTo()

Default constructor used by the command factory.

```
public MoveTo()
```

MoveTo(Canvas, int, int)

Initializes a new instance of the MoveTo command.

```
public MoveTo(Canvas c, int x, int y)
```

Parameters

`c` Canvas

Canvas used for drawing.

`x` int

Target X coordinate.

`y` int

Target Y coordinate.

Fields

x

```
private int x
```

Field Value

int

y

```
private int y
```

Field Value

int

Methods

Execute()

Executes the MoveTo command and moves the drawing cursor.

```
public override void Execute()
```

Exceptions

CanvasException

Thrown when parameters are invalid.

Class Program

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

```
internal static class Program
```

Inheritance

object ← Program

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Methods

Main()

The main entry point for the application.

```
[STAThread]
private static void Main()
```

Class Rectangle

Namespace: BOOSEapp

Assembly: BOOSEapp.dll

BOOSE command that draws a rectangle on the canvas.

```
internal class Rectangle : CommandTwoParameters, ICommand
```

Inheritance

object ← Command ← CanvasCommand ← CommandOneParameter ← CommandTwoParameters ← Rectangle

Implements

ICommand

Inherited Members

CommandTwoParameters.param2 , CommandTwoParameters.param2unprocessed , CommandTwoParameters.CheckParameters(string[]) , CommandOneParameter.param1 , CommandOneParameter.param1unprocessed , CanvasCommand.yPos , CanvasCommand.xPos , CanvasCommand.canvas , CanvasCommand.Canvas , Command.IsDouble , Command.program , Command.parameterList , Command.parameters , Command.paramsint , Command.Set(StoredProgram, string) , Command.Compile() , Command.ProcessParameters(string) , Command.ToString() , Command.Program , Command.Name , Command.ParameterList , Command.Parameters , Command.Paramsint , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object)

Constructors

Rectangle()

Default constructor used by the command factory.

```
public Rectangle()
```

Rectangle(Canvas, int, int)

Initializes a new instance of the Rectangle command.

```
public Rectangle(Canvas c, int width, int height)
```

Parameters

`c` Canvas

Canvas used for drawing.

`width` int

Width of the rectangle.

`height` int

Height of the rectangle.

Fields

height

```
private int height
```

Field Value

int

width

```
private int width
```

Field Value

int

Methods

Execute()

Executes the rectangle command and draws the rectangle.

```
public override void Execute()
```

Exceptions

CanvasException

Thrown when width or height parameters are invalid.

Namespace BOOSEappTests

Classes

CanvasCommandTests

Unit tests for BOOSE canvas commands and interpreter execution.

Class CanvasCommandTests

Namespace: BOOSEappTests

Assembly: BOOSEappTests.dll

Unit tests for BOOSE canvas commands and interpreter execution.

```
[TestClass]
public class CanvasCommandTests
```

Inheritance

object ← CanvasCommandTests

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Fields

_canvas

```
private AppCanvas _canvas
```

Field Value

AppCanvas

_factory

```
private AppCommandFactory _factory
```

Field Value

AppCommandFactory

_parser

```
private IParser _parser
```

Field Value

IParser

_program

```
private StoredProgram _program
```

Field Value

StoredProgram

Methods

DrawToCommand_UpdatesPenPositionVariables()

Tests that the DrawTo command correctly updates the pen position variables after drawing a line.

```
[TestMethod]
public void DrawToCommand_UpdatesPenPositionVariables()
```

MoveToCommand_UpdatesPenPositionVariables()

Tests that the MoveTo command correctly updates the pen position variables (Xpos and Ypos).

```
[TestMethod]
public void MoveToCommand_UpdatesPenPositionVariables()
```

MultilineProgram_ExecutesAndLeavesPenAtExpectedPosition()

Tests execution of a multiline BOOSE program and verifies that the final pen position is correct.

```
[TestMethod]
public void MultilineProgram_ExecutesAndLeavesPenAtExpectedPosition()
```

Setup()

Initializes test environment before each test case. Sets up canvas, factory, stored program and parser.

```
[TestInitialize]
public void Setup()
```